

ASU RRAM Verilog-A Compact Model

1. Introduction

The Verilog-A compact model is developed for metal-oxide based RRAM devices with bipolar switching characteristics. It is improved based on the earlier work originally developed at Stanford [1]. The parameters of the model are re-fitted to the experimental data of IMEC HfO_x-based RRAM devices [2-3]. In the model, the switching mechanism relies on the dynamics of 1D conductive filament growth/dissolution in the oxide layer. For simplicity, there is no definition on the lateral size of the filament. The primary internal variable used in this model is the gap distance (g), which is defined as the distance between the top electrode and the tip of the conductive filament. The gap (g) can determine the RRAM resistance through electron tunneling conduction mechanism. By applying a positive/negative voltage on the RRAM device, the gap will decrease/increase due to the generation/recombination of oxygen vacancies at the tip of conductive filament. Temperature dynamics is also included in this model to emulate the Joule heating effect on the device. This Verilog-A model is compatible with common SPICE simulators, although HSPICE H-2013.03-SP2 is the recommended software. To demonstrate the model, an HSPICE netlist for 1T1R configuration is provided as an example.

Copyright of the model is maintained by the developers, and the model is distributed under the terms of the Creative Commons Attribution-NonCommercial 4.0 International Public License <http://creativecommons.org/licenses/by-nc/4.0/legalcode>. If you use this model in your work, you are requested to cite [1] in the reference.

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Caution:

This RRAM model is a dynamical model that can only be run in transient mode in SPICE, DC mode is not available. If you want to simulate quasi-DC sweep, you should create a staircase voltage pulse trains.

2. Model equations

2.1. I-V relation (electron tunneling)

$$I = I_0 \exp\left(-\frac{g}{g_0}\right) \sinh\left(\frac{V}{V_0}\right) \quad (1)$$

2.2. Local field enhancement factor

$$\gamma = \gamma_0 - \beta \left(\frac{g}{g_1}\right)^3 \quad (2)$$

2.3. Gap dynamics (ion migration)

$$\frac{dg}{dt} = -v_0 \left[\exp\left(-\frac{qE_{ag}}{kT}\right) \exp\left(\frac{\gamma a_0 qV}{L kT}\right) - \exp\left(-\frac{qE_{ar}}{kT}\right) \exp\left(-\frac{\gamma a_0 qV}{L kT}\right) \right] \quad (3)$$

$$g(t+dt) = g(t) + dg \quad (4)$$

2.4. Temperature dynamics

$$\frac{dT}{dt} + \frac{T - T_0}{\tau_{th}} = \frac{|V \times I|}{C_{th}} \quad (5)$$

$$T(t+dt) = T(t) + dT \quad (6)$$

3. Symbols and parameters

3.1. Universal constants

Symbol	Name in code	Unit	Value	Description
k	kb	J/K	1.38e-23	Boltzmann constant
q	q	C	1.6e-19	Elementary unit charge

3.2. Model parameters

Symbol	Name in code	Unit	Default	Range ^a	Description
a_0	a0	m	0.25e-9 ^b	(0:inf)	Atomic hopping distance
E_{ag}	Eag	eV	1.501 ^b	(0:inf)	Activation energy for vacancy generation
E_{ar}	Ear	eV	1.5 ^b	(0:inf)	Activation energy for vacancy recombination
L	L	m	5e-9	(0:inf)	Oxide thickness
g_{min}	gap_min	m	0.1e-9	(0:L)	Minimum gap distance
g_{max}	gap_max	m	1.7e-9	(g_{min} :L)	Maximum gap distance
g_{ini}	gap_ini	m	0.1e-9	[g_{min} : g_{max}]	Initial gap distance
I_0	I0	A	6.14e-5	(0:inf)	I-V fitting parameter
g_0	g0	m	2.75e-10	(0:inf)	I-V fitting parameter
V_0	V0	V	0.43	(0:inf)	I-V fitting parameter
v_0	Vel0	m/s	150	(0:inf)	Gap dynamics fitting parameter
γ_0	gamma0	--	16.5	(0:inf)	Gap dynamics fitting parameter
g_1	g1	m	1e-9	(0:inf)	Gap dynamics fitting parameter
β	beta	--	1.25	$(0: \frac{\gamma_0}{(\frac{g_{max}}{g_1})^3})$	Gap dynamics fitting parameter
T_0	T0	K	298	(0:inf)	Ambient temperature
C_{th}	Cth	J/K	3.18e-16	(0:inf)	Effective thermal capacitance
τ_{th}	Tau_th	s	2.3e-10	(0:inf)	Effective thermal time constant

^a The range is specified to make parameters physically reasonable, or to prevent difficulties in the numerical evaluation, such as division by zero.

^b The values may vary with different material systems of the RRAM device. You may tune these parameters to fit your specific RRAM characteristics.

3.3. Model variables

Symbol	Name in code	Unit	Description
V	V	V	Voltage across the cell
I	I	A	Current flowing through the cell
g	gap	m	Gap distance from the filament tip to the top electrode
dg/dt	gap_ddt	m/s	Gap velocity (growth/dissolution)
γ	gamma	--	Local field enhancement factor
T	temperature	K	Temperature of the conductive filament

4. Gap-R relation

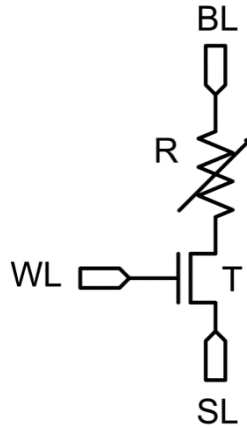
Based on Eq. (1) with the default parameters, the gap-resistance relation is provided below, when the resistance is read at 0.1V. The RRAM can switch within this range depending on the programming condition:

R (Ω)	10k	20k	30k	50k	100k	200k	300k	500k	1M	2M	3M
Gap (nm)	0.1	0.144	0.403	0.543	0.734	0.925	1.036	1.177	1.367	1.558	1.669

5. Model demonstration for 1T1R configuration (SET→RESET in pulse programming mode)

1-transistor-1-resistor (1T1R) memory cell is shown below. The transistor is implemented using PTM model [4] at 65nm technology (model released at February 22, 2006). When the cell is set from high resistance to low resistance state, WL is turned on, and the SET pulse is sent to BL while SL is ground. When the cell is reset from low resistance state to high resistance state, WL is turned on, and the RESET pulse is sent to SL while BL is ground. In this demonstration, both SET and RESET pulse widths are fixed at 10ns. The WL, BL and SL pulse voltages are chosen to achieve the SET resistance switching from $\sim 1\text{M}\Omega$ to $\sim 50\text{k}\Omega$, followed by the RESET resistance switching back to $\sim 1\text{M}\Omega$ again.

Schematic of 1T1R configuration:



The SPICE netlist:

```
1T1R testbench
.option post ingold=1
.include '65nm_bulk.pm'
.hdl 'rram.va'

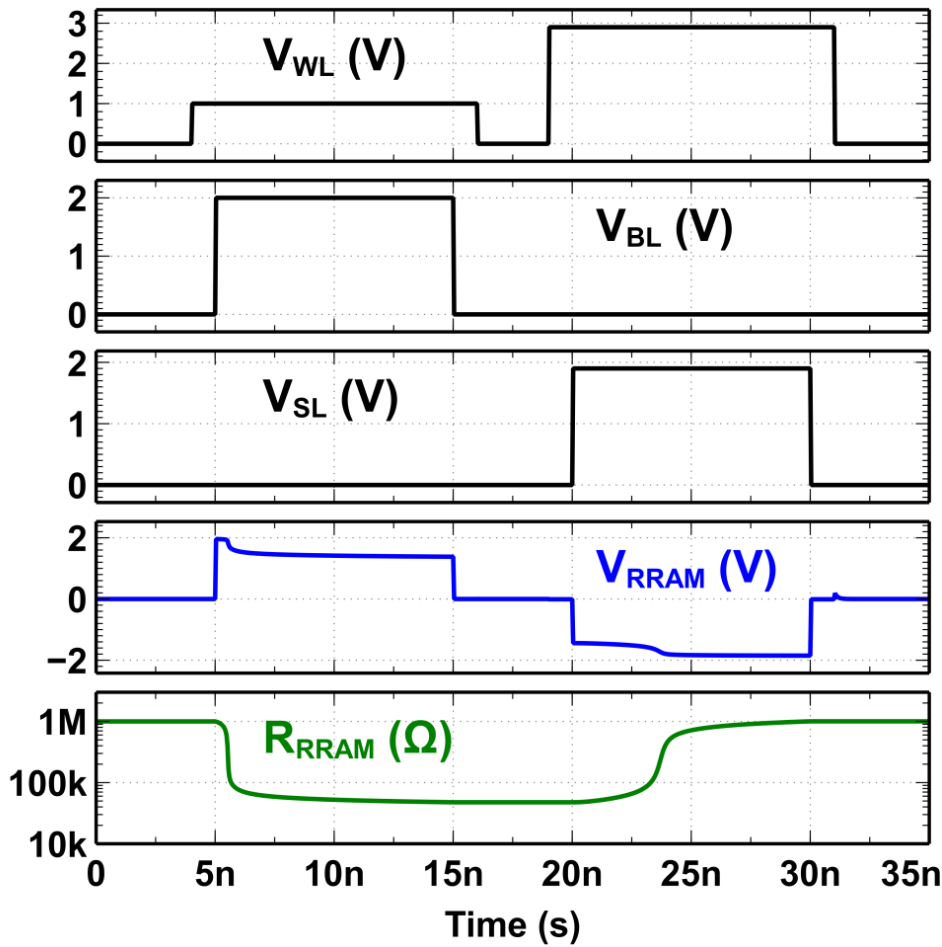
.param tr=10ps          $ Voltage pulse rise time
.param tf=10ps          $ Voltage pulse fall time
.param WL_SET=1.0V      $ WL SET voltage amplitude
.param BL_SET=2.0V      $ BL SET voltage amplitude
.param WL_RESET=2.9V    $ WL RESET voltage amplitude
.param SL_RESET=1.9V    $ SL RESET voltage amplitude
.param SET_PULSE=10ns   $ SET pulse width
.param SET_START=5ns    $ SET operation delay
.param RESET_PULSE=10ns $ RESET pulse width
.param RESET_START='SET_START+5n+SET_PULSE' $ RESET operation delay
.param T_FINAL='RESET_START+RESET_PULSE+5n' $ Final simulation time

X1 BL Nb rram gap_ini=1.367n tstep=10ps $ RRAM
M1 Nb WL SL 0 nmos w=130n l=65n $ Transistor

V_WL_SET WL WL2 PULSE (0V WL_SET 'SET_START-1n' tr tf 'SET_PULSE+2n' T_FINAL)
V_WL_RESET WL2 0 PULSE (0V WL_RESET 'RESET_START-1n' tr tf 'RESET_PULSE+2n' T_FINAL)
V_BL_SET BL 0 PULSE (0V BL_SET 'SET_START' tr tf SET_PULSE T_FINAL)
V_SL_RESET SL 0 PULSE (0V SL_RESET 'RESET_START' tr tf RESET_PULSE T_FINAL)

.tran 40ps T_FINAL
.print V(WL) V(BL) V(SL) V(BL,Nb)
.print V(X1.gap_out) V(X1.R_out)
.end
```

Simulation results by HSPICE 2013:



References:

- [1] P.-Y. Chen, S. Yu, "Compact modeling of RRAM devices and its applications in 1T1R and 1S1R array design," *IEEE Transactions on Electron Devices*, vol. 62, no. 12, pp. 4022-4028, 2015.
- [2] Y. Y. Chen, B. Govoreanu, L. Goux, R. Degraeve, A. Fantini, G. S. Kar, D. J. Wouters, G. Groeseneken, J. A. Kittl, M. Jurczak, and L. Altimime, "Balancing SET/RESET pulse for endurance in 1T1R bipolar RRAM," *IEEE Transactions on Electron Devices*, vol. 59, pp. 3243-3249, 2012.
- [3] Y. Y. Chen, M. Komura, R. Degraeve, B. Govoreanu, L. Goux, A. Fantini, N. Raghavan, S. Clima, L. Zhang, A. Belmonte, A. Redolfi, G. S. Kar, G. Groeseneken, J. A. Kittl and M. Jurczak, "Improvement of data retention in HfO₂/Hf 1T1R RRAM cell under low operating current," *IEEE International Electron Devices Meeting (IEDM)*, pp. 252-255, 2013.
- [4] Predictive Technology Model (PTM), <http://ptm.asu.edu/>